

SUBPOWER ENERGY AND DYADIC DEFECT CRITERIA FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. Let

$$b_n = -\mu(n) \log n, \quad B_N = \sum_{n \leq N} b_n.$$

We prove that the Riemann hypothesis is equivalent to subpower growth of the positive prefix energy

$$\sum_{n \leq X} \frac{B_n^2}{n(n+1)},$$

and also to convergence of each positive power-damped version of this sum. The same criterion has a continuous form: for every $\sigma > 1/2$, the weighted energy of the right-continuous step function $B(x) = B_{\lfloor x \rfloor}$ is finite, or equivalently its Mellin transform is square integrable on $\Re s = \sigma$. An exact prefix-square identity then gives equivalent cumulative and individual lower bounds for signed dyadic defects. As a stronger sufficient condition, eventual nonnegativity of the dyadic defects implies the Riemann hypothesis, simplicity of all nontrivial zeros, and the explicit bound

$$\sum_{\rho} \left| \frac{1}{\rho \zeta'(\rho)} \right|^2 \leq \frac{18}{\pi^2}.$$

The paper proves these equivalences, but it does not establish any of the equivalent estimates or the eventual-sign hypothesis. It therefore does not prove the Riemann hypothesis.

1. INTRODUCTION

The classical Littlewood criterion expresses the Riemann hypothesis through cancellation in the summatory Möbius function. This paper studies the closely related coefficients

$$b_n = -\mu(n) \log n = (\mu * \Lambda)(n),$$

which are the Dirichlet coefficients of the derivative of $1/\zeta(s)$. Their prefix sums B_N lead to two views of the same criterion.

The first view is positive. It asks whether the cumulative square energy

$$A(X) = \sum_{2 \leq n \leq X} \frac{B_n^2}{n(n+1)}$$

grows more slowly than every fixed positive power of X . Positivity makes this condition stable under damping and lets it pass directly to a weighted Mellin L^2 norm.

The second view is signed and local. On the block $(L, R]$, define

$$D(L, R) = \sum_{L < n \leq R} \frac{b_n^2 - b_n B_{n-1}}{n}.$$

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An exact identity relates $D(L, R)$ to the positive prefix energy. On the power-of-two blocks $(2^j, 2^{j+1}]$, this identity shows that subpower lower bounds for either each defect or their cumulative sum are equivalent to the Riemann hypothesis.

The principal theorem is an equivalence theorem, not an estimate for the new quantities. It does not prove any of its criteria. A separate corollary starts from the stronger, also unproved, hypothesis that the power-of-two defects are eventually nonnegative. That sign condition supplies a sharp leading constant in the prefix energy. Mellin-Plancherel then yields an explicit boundary norm. The poles of the differentiated reciprocal zeta function consume nonnegative portions of this norm, which forces simple zeros and the stated reciprocal-residue budget.

Several L^2 formulations of the Riemann hypothesis are already known. The Nyman-Beurling theory and its refinements use approximation or closure of families of fractional-part functions [8, 4, 1, 6]. The weak Mertens theory uses mean squares of the ordinary Mertens function and has strong consequences for reciprocal zeta residues [9, 7]. Modified mollifiers, Möbius-convolution criteria, generalized von Mangoldt sums, and smoothed Möbius transforms provide further nearby formulations [5, 2, 3, 10]. The criterion here instead uses the literal prefix of $-\mu(n) \log n$, its positive weighted square energy, and an exact signed dyadic identity. These comparisons locate the argument; they do not establish a novelty or priority claim.

2. THE EXACT BLOCK IDENTITY

Let μ be the Möbius function and Λ the von Mangoldt function. For $n \geq 1$, set

$$(1) \quad b_n = (\mu * \Lambda)(n) = -\mu(n) \log n, \quad B_N = \sum_{n \leq N} b_n.$$

The coefficient identity follows by differentiating $1/\zeta(s) = \sum_{n \geq 1} \mu(n)n^{-s}$ in $\Re s > 1$. We also use the right-continuous step function

$$(2) \quad B(x) = \sum_{n \leq x} b_n = B_{[x]}, \quad x \geq 1.$$

Values at the integer endpoints do not affect the integrals below, but this convention fixes the discrete-to-continuous correspondence.

For integers $2 \leq L < R$, define

$$(3) \quad D(L, R) = \sum_{L < n \leq R} \frac{b_n^2 - b_n B_{n-1}}{n},$$

$$(4) \quad V(L, R) = \sum_{L < n \leq R} \frac{\mu(n)^2 \log^2 n}{n}.$$

Proposition 2.1 (Prefix-square identity). *For every $2 \leq L < R$,*

$$(5) \quad 2D(L, R) = 3V(L, R) - H(L, R),$$

where

$$(6) \quad H(L, R) = \frac{B_R^2}{R} - \frac{B_L^2}{L+1} + \sum_{L < n < R} \frac{B_n^2}{n(n+1)}.$$

Proof. Since $b_n = B_n - B_{n-1}$,

$$2b_n B_{n-1} = B_n^2 - B_{n-1}^2 - b_n^2.$$

It follows that

$$2(b_n^2 - b_n B_{n-1}) = 3b_n^2 - B_n^2 + B_{n-1}^2.$$

After division by n , summation over $L < n \leq R$ gives

$$\sum_{L < n \leq R} \frac{B_n^2 - B_{n-1}^2}{n} = \frac{B_R^2}{R} - \frac{B_L^2}{L+1} + \sum_{L < n < R} \frac{B_n^2}{n(n+1)}.$$

Now use $b_n^2 = \mu(n)^2 \log^2 n$. □

For $j \geq 1$, write

$$(7) \quad D_j = D(2^j, 2^{j+1}), \quad S_J = \sum_{j=1}^{J-1} D_j \quad (J \geq 2).$$

Summing Proposition 2.1 over adjacent dyadic blocks gives the following exact global form.

Corollary 2.2 (Exact dyadic balance). *For $X = 2^J$ with $J \geq 2$,*

$$(8) \quad 2S_J + A(X) + \frac{B_X^2}{X+1} = 3W(X) + \frac{B_2^2}{2},$$

where

$$W(X) = \sum_{2 < n \leq X} \frac{\mu(n)^2 \log^2 n}{n}.$$

Proof. At an interior dyadic boundary n , the adjacent endpoint terms combine as

$$\frac{B_n^2}{n} - \frac{B_n^2}{n+1} = \frac{B_n^2}{n(n+1)}.$$

$$\sum_{j=1}^{J-1} H(2^j, 2^{j+1}) = \frac{B_X^2}{X} - \frac{B_2^2}{3} + \sum_{2 < n < X} \frac{B_n^2}{n(n+1)}.$$

Substitution in Proposition 2.1, followed by

$$A(X) = \frac{B_2^2}{6} + \sum_{2 < n < X} \frac{B_n^2}{n(n+1)} + \frac{B_X^2}{X(X+1)},$$

gives (8). □

We shall use the standard squarefree estimate and its partial-summation consequence

$$(9) \quad \sum_{n \leq X} \mu(n)^2 = \frac{6}{\pi^2} X + O(\sqrt{X}),$$

$$(10) \quad \sum_{n \leq X} \frac{\mu(n)^2 \log^2 n}{n} = \frac{2}{\pi^2} \log^3 X + O(\log^2 X).$$

3. POSITIVE ENERGY, MELLIN ENERGY, AND DYADIC DEFECTS

Define the positive cumulative energy

$$(11) \quad A(X) = \sum_{2 \leq n \leq X} \frac{B_n^2}{n(n+1)}.$$

For $\eta > 0$ and $\sigma > 1/2$, also define

$$(12) \quad \mathcal{E}_\eta = \sum_{n \geq 2} \frac{B_n^2}{n(n+1)} n^{-\eta},$$

$$(13) \quad I(\sigma) = \int_1^\infty |B(x)|^2 x^{-2\sigma-1} dx.$$

Lemma 3.1 (Exact continuous realization). *For every $\sigma > 1/2$,*

$$(14) \quad I(\sigma) = \sum_{n \geq 2} B_n^2 \frac{n^{-2\sigma} - (n+1)^{-2\sigma}}{2\sigma}.$$

If $\eta = 2\sigma - 1$, then

$$(15) \quad I(\sigma) < \infty \iff \mathcal{E}_\eta < \infty.$$

Put

$$g_\sigma(y) = B(e^y) e^{-\sigma y} \mathbf{1}_{[0, \infty)}(y)$$

and, for $T > 0$, let

$$F_{\sigma, T}(t) = \int_0^T g_\sigma(y) e^{-ity} dy.$$

Then $I(\sigma) < \infty$ if and only if $F_{\sigma, T}$ converges in $L^2(\mathbb{R})$ as $T \rightarrow \infty$. Denote the limit by F_σ . In that case,

$$(16) \quad \int_{-\infty}^{\infty} |F_\sigma(t)|^2 dt = 2\pi I(\sigma).$$

Proof. By (2), $B(x) = B_n$ on $[n, n+1)$ apart from an irrelevant endpoint. Integration on each interval gives (14). The mean-value theorem gives

$$\frac{n^{-2\sigma} - (n+1)^{-2\sigma}}{2\sigma} \asymp_\sigma n^{-2\sigma-1} = n^{-2-\eta} \asymp \frac{n^{-\eta}}{n(n+1)},$$

which proves (15). The substitution $x = e^y$ gives

$$\int_0^\infty |g_\sigma(y)|^2 dy = I(\sigma).$$

For $T > U$, Plancherel for the convention $\widehat{g}(t) = \int_{\mathbb{R}} g(y) e^{-ity} dy$ gives

$$\|F_{\sigma, T} - F_{\sigma, U}\|_2^2 = 2\pi \int_U^T |g_\sigma(y)|^2 dy.$$

Thus $F_{\sigma, T}$ is Cauchy in L^2 exactly when $I(\sigma)$ is finite. Taking the limit proves (16). $\square$

Theorem 3.2 (Equivalent subpower criteria). *The following statements are equivalent.*

- (1) *The Riemann hypothesis is true.*
- (2) *For every $\varepsilon > 0$,*

$$(17) \quad A(X) = O_\varepsilon(X^\varepsilon).$$

- (3) *For every $\eta > 0$, the series $\mathcal{E}_\eta$ converges.*
- (4) *For every $\sigma > 1/2$, the integral $I(\sigma)$ is finite.*
- (5) *For every $\sigma > 1/2$, the truncated transforms $F_{\sigma, T}$ in Lemma 3.1 converge in $L^2(\mathbb{R})$.*
- (6) *For every $\varepsilon > 0$, there is C_ε such that*

$$(18) \quad S_J \geq -C_\varepsilon 2^{\varepsilon J} \quad (J \geq 2).$$

- (7) *For every $\varepsilon > 0$, there is C_ε such that*

$$(19) \quad D_j \geq -C_\varepsilon 2^{\varepsilon j} \quad (j \geq 1).$$

Proof. We first connect the positive discrete conditions. If (17) holds and $\eta > 0$, use the bound with exponent $\eta/2$ and partial summation against the decreasing weight $x^{-\eta}$; this proves $\mathcal{E}_\eta < \infty$. Conversely, positivity gives

$$A(X) \leq X^\eta \mathcal{E}_\eta.$$

Taking $\eta = \varepsilon$ proves (17). Lemma 3.1 then proves the equivalence of statements 2 through 5.

We next show that statement 3 implies RH. On $\Re s > 1$, partial summation gives

$$(20) \quad F(s) := \int_1^\infty B(x)x^{-s-1} dx = \frac{1}{s} \left(\frac{1}{\zeta(s)} \right)'.$$

Let K be a compact subset of $\Re s > 1/2$, and choose σ_0 with $1/2 < \sigma_0 < \min_{s \in K} \Re s$. Statement 3 and Lemma 3.1 give $I(\sigma_0) < \infty$. Cauchy-Schwarz yields

$$\int_1^\infty |B(x)|x^{-\Re s-1} dx \leq I(\sigma_0)^{1/2} \left(\int_1^\infty x^{-2(\Re s-\sigma_0)-1} dx \right)^{1/2},$$

uniformly on K . Thus the integral defining F is locally uniformly convergent and holomorphic in $\Re s > 1/2$.

The half-plane is simply connected, so $sF(s)$ has a holomorphic primitive $Q(s)$ there. Normalize it at one point in $\Re s > 1$ to equal $1/\zeta(s)$. In $\Re s > 1$, equation (20) gives $Q' = (1/\zeta)'$, so the normalization gives $Q = 1/\zeta$. On $\{\Re s > 1/2\} \setminus \{1\}$, the meromorphic identity theorem gives

$$\zeta(s)Q(s) = 1.$$

Hence ζ has no zero with real part greater than $1/2$. The functional equation reflects every nontrivial zero across the critical line, so all nontrivial zeros lie on that line. This proves RH. Moreover, $Q' = sF$ and $Q = 1/\zeta$ give

$$F(s) = \frac{1}{s} \left(\frac{1}{\zeta(s)} \right)' \quad (\Re s > 1/2, s \neq 1).$$

Both sides extend holomorphically across $s = 1$. The same meromorphic expression therefore controls the approach from the right to a zero on the critical line.

Conversely, RH implies the classical estimate [9, Theorem 14.25(C)]

$$M(x) := \sum_{n \leq x} \mu(n) = O_\alpha(x^{1/2+\alpha})$$

for every $\alpha > 0$. Partial summation in (1) gives

$$(21) \quad B(x) = -M(x) \log x + \int_1^x \frac{M(t)}{t} dt.$$

After absorbing the logarithm into a slightly larger exponent, $B(x) = O_\alpha(x^{1/2+\alpha})$ for every $\alpha > 0$. Substitution in (11) proves (17).

It remains to connect the dyadic statements. Statement 7 implies statement 6 by summing a geometric progression. If statement 6 holds, then Corollary 2.2 and (10) give, for $X = 2^j$,

$$A(X) \leq 3W(X) + 2C_\varepsilon X^\varepsilon + O(1) = O_\varepsilon(X^\varepsilon).$$

Monotonicity of A extends this bound to every X , so statement 2 follows.

Finally assume RH. Choose $\alpha > 0$ so small that $2\alpha < \varepsilon$. The bound after (21) and (6) give

$$H(2^j, 2^{j+1}) = O_\alpha(2^{2\alpha j}).$$

Since $V \geq 0$, Proposition 2.1 yields

$$D_j \geq -O_\alpha(2^{2\alpha j}),$$

which is statement 7. All seven statements are therefore equivalent. $\square$

Remark 3.3 (Boundary not included). The Mellin conditions in Theorem 3.2 concern every fixed line $\Re s = \sigma > 1/2$. The theorem does not assert square integrability on the critical line itself.

4. THE STRONGER EVENTUAL-SIGN CONDITION

The equivalence theorem allows small negative dyadic defects of subpower size. Eventual nonnegativity is stronger and supplies an explicit leading constant.

Theorem 4.1 (Eventual sign and boundary budget). *Suppose there is j_0 such that*

$$(22) \quad D_j \geq 0 \quad (j \geq j_0).$$

Then the Riemann hypothesis is true, every nontrivial zero of ζ is simple, and

$$(23) \quad \sum_{\rho} \left| \frac{1}{\rho \zeta'(\rho)} \right|^2 \leq \frac{18}{\pi^2}.$$

The sum includes both conjugate ordinates. In addition,

$$(24) \quad A(X) \leq \frac{6}{\pi^2} \log^3 X + O(\log^2 X)$$

and, with F as in (20),

$$(25) \quad \limsup_{\delta \downarrow 0} \delta^3 \int_{-\infty}^{\infty} \left| F\left(\frac{1}{2} + \delta + it\right) \right|^2 dt \leq \frac{9}{\pi}.$$

Proof. Condition (22) and Corollary 2.2 give

$$A(2^J) \leq 3W(2^J) + O(1).$$

Equation (10) proves (24) first at powers of two and then, by monotonicity, for all X . Theorem 3.2 now proves RH.

For $\delta > 0$, the step convention gives

$$\begin{aligned} I\left(\frac{1}{2} + \delta\right) &= \sum_{n \geq 2} B_n^2 \int_n^{n+1} x^{-2-2\delta} dx \\ &\leq \sum_{n \geq 2} \frac{B_n^2}{n(n+1)} n^{-2\delta} = \int_{2^-}^{\infty} x^{-2\delta} dA(x). \end{aligned}$$

Integration by parts and (24) show that

$$(26) \quad \limsup_{\delta \downarrow 0} \delta^3 I\left(\frac{1}{2} + \delta\right) \leq \frac{9}{2\pi^2}.$$

Indeed,

$$\frac{6}{\pi^2} 2\delta \int_1^{\infty} \log^3 x x^{-2\delta-1} dx = \frac{9}{2\pi^2 \delta^3},$$

while the $O(\log^2 x)$ term contributes $O(\delta^{-2})$. Equation (16) turns (26) into (25).

Let $\rho = 1/2 + i\gamma$ be a nontrivial zero of multiplicity m , and write

$$\zeta(s) = (s - \rho)^m g(s), \quad g(\rho) \neq 0.$$

In a right-hand neighborhood of ρ , the preceding meromorphic identity gives

$$(27) \quad F(s) = c_{\rho}(s - \rho)^{-m-1} + O_{\rho}(|s - \rho|^{-m}), \quad c_{\rho} = -\frac{m}{\rho g(\rho)}.$$

Choose $\eta > 0$ so that the Laurent estimate holds in the right half of $|s - \rho| < 2\eta$, the disk avoids 0 and 1, and ρ is its only zeta zero. Put $I_{\rho} = [\gamma - \eta, \gamma + \eta]$. Scaling $t - \gamma = \delta u$ in (27) gives

$$(28) \quad \int_{I_{\rho}} |F(1/2 + \delta + it)|^2 dt = |c_{\rho}|^2 C_m \delta^{-2m-1} + o_{\rho}(\delta^{-2m-1}),$$

where

$$C_m = \int_{-\infty}^{\infty} (1 + u^2)^{-m-1} du > 0.$$

Indeed, the squared remainder in (27) contributes $O_\rho(\delta^{1-2m})$, and the cross term contributes $O_\rho(\delta^{-2m})$. If $m \geq 2$, multiplying (28) by δ^3 makes its left side tend to infinity, contradicting the finite global limsup in (25). Thus every nontrivial zero is simple.

For $m = 1$, $g(\rho) = \zeta'(\rho)$ and

$$C_1 = \int_{-\infty}^{\infty} \frac{du}{(1+u^2)^2} = \frac{\pi}{2}.$$

Consequently, (28) becomes

$$(29) \quad \int_{I_\rho} |F(1/2 + \delta + it)|^2 dt = \frac{\pi}{2\delta^3} \left| \frac{1}{\rho\zeta'(\rho)} \right|^2 + o_\rho(\delta^{-3}).$$

Fix a finite set E of zeros and choose the intervals I_ρ pairwise disjoint. The global norm is nonnegative. Summing (29) over E and using (25) gives

$$\frac{\pi}{2} \sum_{\rho \in E} \left| \frac{1}{\rho\zeta'(\rho)} \right|^2 \leq \frac{9}{\pi}.$$

Taking the supremum over all finite E proves (23). $\square$

5. RELATION TO THE P1 CONDITION

A previously considered condition asks for more than (22). For every integer $L \geq 2$ and $R \in \{2L, 2L+1\}$, let $D(L, R)$ be as in (3).

Definition 5.1 (P1). Condition P1 asserts

$$D(L, R) \geq 0$$

for every $L \geq 2$ and both permitted endpoints $R = 2L$ and $R = 2L+1$.

Taking $L = 2^j$ shows that P1 implies (22), with $j_0 = 1$. Theorem 4.1 therefore gives

$$\text{P1} \implies \text{RH, simple nontrivial zeros, and (23)}.$$

This is only a one-way implication. Theorem 3.2 proves that the subpower energy and defect bounds are equivalent to RH, but it does not prove that RH implies P1. It also does not prove a strict logical separation between P1 and RH. Condition P1 remains unproved, and finite verification of its inequalities cannot establish the universal statement.

6. CONTEXT AND LIMITATIONS

The Mellin map in Lemma 3.1 places the positive energy criterion in a classical analytic setting, but its role differs from the Nyman-Beurling program. Nyman and Beurling characterize RH by density of specified fractional-part functions in an L^2 space [8, 4]. Báez-Duarte restricts the admissible generators to an arithmetic family [1]. Kapustin gives unitarily equivalent Hardy-space and weighted- L^2 approximation models, including a closure criterion for the span of $\rho(nx)$ [6, Theorem 1]. Here there is no approximation family: the tested object is the single step function formed by the literal prefixes of $-\mu(n) \log n$, and the condition is finiteness of its weighted square energy on every line to the right of the critical line.

The weak Mertens condition also uses a positive mean square and implies RH, simple zeros, and summability of reciprocal residues [9, 7]. Its summatory function is $M(x) = \sum_{n \leq x} \mu(n)$, whereas the present energy uses the logarithmic prefix $B(x)$ and arises from $(1/\zeta)'$. The explicit constant in Theorem 4.1 comes from the exact dyadic identity and the leading squarefree density.

The criteria of Conrey and Myerson use modified Möbius mollifiers in the Balazard-Saias framework [5]; Báez-Duarte develops general Möbius-convolution criteria [2]; Banks and Sinha use uniform twisted sums of generalized von Mangoldt functions [3]; and Verjovsky

studies exponential smoothing and local Fourier formulations of Möbius cancellation [10]. These works share coefficients or transform methods with the present paper, but their stated criteria are different.

No computation is used in any proof above. In particular, finite tests of the signed defects and finite partial sums of (23) cannot prove the universal hypotheses or control their tails. The paper establishes equivalences and conditional consequences only.

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